

MEC - 107 INTERNATIONAL TRADE & DEVELOPMENT



INTERNATIONAL TRADE AND DEVELOPMENT

(Vol. 2)



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Unit 4	Alternative Explanations of Trade	Dr. Pravin Jadhav, Assistant Professor, Institute of Infrastructure Technology Research and Management (IITRAM)
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Unit 17	Trade and Environment	Prof. S.N. Mukherjee, National Institute of Public Finance and Policy
BLOCK 8	INDIA'S FOREIGN TRADE: POLICY AND CHALLENGES	
Unit 18	India's Trade Policy	Ashish Gupta, Motilal Nehru College, Delhi University
Unit 19	India's Trade: Trends, Composition and Challenges	Dr. Pravin Jadhav, Assistant Professor, Institute of Infrastructure Technology Research and Management (IITRAM)



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COURSE INTRODUCTION

The global economy is constantly evolving, shaped by technological advancements, geopolitical shifts, and changing trade policies. The process of globalization has interconnected economies around the world. Increased cross-border trade, investment, and the international mobility of capital, labour, and technology have facilitated the integration of national economies into the global marketplace. Further, geopolitical dynamics and power shifts among nations have profound effects on cross-border flows and global economic integration. In light of the ever-changing, dynamic nature of the global economy, the course **MEC 107 ‘International Trade and Development’** aims to understand these drivers of international (both inter and intra-industry) trade and international finance.

This course will provide you with valuable knowledge and skills to understand and navigate the complexities of the global economy. International trade plays a crucial role in the economic development of nations. By studying this course, learners will gain a deep understanding of the factors that influence global trade patterns, such as trade policies, tariffs, exchange rates, and trade agreements. Geopolitical factors play a crucial role in shaping the global economy. Political stability, regional conflicts, and international relations impact trade flows, investment patterns, and economic alliances. Studying international trade and finance will equip the learners with the knowledge to analyze geopolitical developments and their implications on international business.

The course will also throw light into the debate of free trade versus protectionism by discussing what happens when countries adopt various instruments of protectionism and how the world is affected. In the light of changing nature of agreements between the countries and the new world order, the course also delves into various types of trade agreements and India's agreements with various nations. It also digs deeper into the evolved role of India in WTO and traces the evolving role since the GATT era. India's foreign trade policy, how it has evolved and what are the challenges associated with it are also discussed in depth. The course will equip the learners with an understanding of the impact of various policies of countries and international organisations that affect global patterns of trade and finance.

The course is structured around eight blocks. Each block is further subdivided into several units. Each Unit is self-contained and has linkages with all other Units.

Block 1: Theory of International Trade introduces the course by discussing the rationale behind International trade and why countries engage in trade. This block examines various theories of International trade and presents their comparative analysis. It identifies the various gains associated with international trade and how they can be measured. It also presents the comparative analysis of various types of gains in this block. As we move forward, we discuss the complementary theories of international trade that are formed by dropping the classical assumptions.

In **Block 2: Modern Theories of International Trade**. This block also delves into intra industry trade (IIT) i.e. the exchange of similar or closely related goods and services within the same

industry between countries or regions. IIT arises due to product differentiation and the diversity of consumer preferences is often driven by economies of scale and the ability to achieve cost efficiencies through mass production. We discuss the need, rationale and measurement of Intra-Industry Trade (IIT) in this block.

After discussing the need and theories of inter-industry and intra-industry trade, we present the two contrasting approaches to international trade: Free trade and protectionism, representing different policy stances and perspectives on the regulation of trade flows in **Block 3: Free trade versus protectionism**. We discuss the latest developments in India's Protectionism Policy and their implications for the Indian economy. To understand how countries adopt free trade and protectionist policies based on their specific economic and political considerations, we also discuss various instruments of protectionism-tariff and non-tariff barriers in this block. We examine their impact on the world economy as well.

As we move to **Block 4: Exchange Rate and Balance of Payments**, we examine the vital role that exchange rates and the Balance of Payments play in a country's macroeconomic stability. Exchange rate policies have significant implications for a country's trade competitiveness, economic growth, and financial stability. Some countries adopt fixed exchange rate systems, while others have floating or managed exchange rate regimes. In this block, we study exchange rates and their management to comprehend the advantages, disadvantages, and challenges associated with different exchange rate systems. We also trace the historical development of international exchange rate regimes and examine operational aspects of the current exchange rate regimes. We discuss the balance of payment disequilibrium and the various approaches that countries adopt for BOP adjustment. We also analyse the experience of various countries to understand the Macroeconomic policy trilemma, the impossible trinity.

International finance is closely intertwined with international trade. **Block 5: International Finance** will help to gain an understanding of global financial markets, capital flows, and international investment. Block 5 gives an idea of the globalization of financial markets and then then the concept of International financial markets. It throws light on the global financial crisis of the year 2008. To understand this crisis better, some of the most devastating financial crises in history are also presented which will add to the understanding of the unfolding of the 2008 crisis. The block also summarises initial monetary and fiscal policy responses to manage this global crisis.

Since Globalization has opened immense opportunities for businesses to expand beyond national borders, we deal in depth with globalization in **Block 6: Globalisation**. The role of the World Trade Organization (WTO) and other multilateral organizations is becoming increasingly important as the world has become more interconnected than ever before, with increased cross-border trade, investment, and economic integration. This block furnishes an understanding of the history and present status of the multilateral trading system in terms of GATT, and WTO rounds. We also discuss the types of regional trading agreements, preferential trading arrangements, free trade areas and customs unions. India's role has also evolved in the multilateral trading systems and the agreements of which India is an integral part e.g., the Asia-Pacific Trade agreement,

ASEAN-India free trade area, SAFTA. The block also presents the contribution of India to WTO and how India has transformed itself with the change of GATT towards the WTO regime.

In **Block 7: Trade, Growth and Development**, we describe the relationship between economic growth and trade. The block presents the various trade policy regimes vis-à-vis growth in developing countries wherein the import substitution and outward-oriented development strategies are discussed. The block also discusses India's experience with inward-oriented and outward-oriented regimes. Because of the increased concern about climate change, we also discuss the link of trade with climate change and the policies and multilateral environmental agreements concerning trade.

Block 8: India's Foreign Trade: Policy and Challenges digs into the concepts, nature and aims of trade policy and the basic tools of a trade policy. It will further explain the evolution of the trade policy in India. It also presents the pattern, direction and composition of India's foreign trade. The challenges faced by the foreign trade of India be highlighted in the concluding part of the block.





BLOCK 5

INTERNATIONAL FINANCE

Unit 11	International Financial Markets and Instruments
Unit 12	Financial and Currency Crises

BLOCK INTRODUCTION

This block is on “**INTERNATIONAL FINANCE**”. It has two units i.e., Unit 11 - **International Financial Markets and Instruments** and Unit 12 - **Financial and Currency Crises**.

Unit 11 provides an idea of globalization of financial markets and then presents the concept of International financial markets. The financial instruments available in the international financial markets are also discussed. The learner also learns about the types and importance of international financial markets. Various instruments (financial as well as debt instruments) which help smooth functioning of the international financial markets are also highlighted.

Unit 12 provides an understanding of global financial crisis of the year 2008. To understand this crisis better, some of the most devastating financial crises in history are also presented which will add to the understanding of the unfolding of the 2008 crisis. It becomes imperative for the learner to understand the impact and effect of such a global crisis on the macroeconomy. The unit completes by presenting the initial policy responses in terms of monetary and fiscal policies, to combat this global crisis.

UNIT 11 INTERNATIONAL FINANCIAL MARKETS AND INSTRUMENTS *

Structure

- 11.0 Objectives
- 11.1 Introduction
- 11.2 Globalisation of Financial Markets
- 11.3 Concept of International Financial Markets
 - 11.3.1 Concept of International Financial Instruments
- 11.4 Types of International Financial Markets
- 11.5 Importance of International Financial Markets and Instruments
- 11.6 Instruments of International Financial Markets
 - 11.6.1 Financial Instruments
 - 11.6.2 International Debt Instruments
- 11.7 Foreign Exchange Exposure/Risk
- 11.8 Let Us Sum Up
- 11.9 Key Words
- 11.10 Some Useful references
- 11.11 Answer/Hints to Check Your Progress Exercises

11.0 OBJECTIVES

After studying this Unit, you should be able to:

- Explain the concepts of International Financial Markets and Instruments;
- Indicate the importance of International Financial Markets;
- Identify the participants in International Financial Markets;
- Distinguish between various instruments of International Financial Markets;
- Describe various International Financial Instruments used for raising finance;
and
- Analyse different types of foreign exchange risk.

* Contributed by Dr. Surendra Kumar, PGDAV College, Delhi University

11.1 INTRODUCTION

Even the most cursory glance at the international economic trends since the 1980s shows revolutionary changes in world financial markets. The interaction of several factors has produced huge capital flows across national boundaries. Along with this, the structure and operation of global financial markets have been transformed.

Financial markets describe where buyers and sellers participate in the trade of assets such as equities, bonds, currencies and derivatives. In such markets, companies get capital to expand, as in the stock or bond markets, or hedge their risks, like the foreign exchange or commodities markets.

The immense growth and spread of international financial markets and because of their ups and downs affect the lives of people not only during times of crisis but also daily. With globalization, deregulation, liberalization, financial and technological innovation, the volume of financial transactions increased to astronomical figures.

The financial market is a highly complex system in which rules, individuals and institutions interact. This complexity increases even more in time and space, particularly in the case of international financial markets. This phenomenon is so pervasive that it created great interest among people in finance and fluctuations in financial markets.

Because of substantial international capital movements, policies and developments in other countries increasingly influence domestic economic performance. With this backdrop, this unit is devoted to the discussion of globalization of financial markets, concepts, functions and importance of international financial markets, instruments used for raising finance from various entities, and their description and analysis of foreign exchange risks.

11.2 GLOBALISATION OF FINANCIAL MARKETS

In the contemporary world, there has been globalisation of financial markets. In a global financial market, technological advances in payment, settlement and trading systems and financial information systems can be made instantaneously available to all market participants. In financial markets, globalisation opens access to borrowing, lending and investing worldwide. In addition, it plays a central role in disciplining policies. This trend is very much evident from the following development (Otmars, 2000):

- i) There has been a steady increase in cross-border financial flows around the world. Because various banks and institutional investors have expanded their activities geographically. In this process, they acted as intermediaries to channel funds from lenders to borrowers across national borders.

- ii) The mature securities markets have gained cross-border orientation as they design and offer securities which attract international investors.
- iii) Since the 1980s, most countries have been following the policy of liberalization, which led to the progressive dismantling of controls on cross-border financial flows.
- iv) Throughout the world, borrowers and lenders find a wide range of funds available for financing deficits and managing surpluses.
- v) Both technological advances and financial innovations are important contributing factors as information systems have become able to compute and store huge amount of data more rapidly.
- vi) More reliable data exchange protocols have made it possible to connect computing machines in more efficient ways. This development facilitated cross-border financial deals both easier and more secure. Besides, it removes the distance problem created by geography or other factors.
- vii) There has been innovation of financial instruments like "derivative" which has made it possible for borrowers and lenders to customise their risk exposures as well as adjust them over time. This is particularly important for international financial markets where information asymmetries exist.

Globalisation is a process that is not yet finalised and will lead to a world in which countries and economic areas become increasingly interdependent. As such, Otmar (2000) remarked that "the globalisation of financial markets is better seen as a process rather than as a fixed state of affairs". It is also important to note that global financial markets are not free from crises as is evident from 1997 East Asian crises, US crisis of 2008 etc. There have been numerous occurrences of financial crises. Some of these financial crises exerted disruptive and protracted effects on the economy locally and globally. As a result, economic policymakers and market participants have devoted considerable attention to the issue of how best to avoid financial crises.

Similarly, very limited room for maneuver is available for economic policy pursuing specific objectives at the national level. This calls for overhauling the architecture of the international financial system.

11.3 CONCEPT OF INTERNATIONAL FINANCIAL MARKETS

The international financial markets consist of sources of finance and uses for finance in several different countries. Each of these is a capital market on its own. The international financial markets are markets where different entities (individuals, financial and non-financial institutions) buy and sell foreign assets and assets are traded between agents in surplus and agents in deficit. The financial market comprises the markets for stock market, bond market, currency

market, derivatives market, commodity market and money market. International institutions lay down rules for the conduct of financial markets. Borrowers/Issuers, Lenders/Investors and Intermediaries are the major players of the international markets. Borrowers/Issuers are corporates, banks, financial institutions, government and quasigovernment bodies and supranational organizations, which need Forex funds for various reasons. The institutions which work in financial markets have different aims and functions like Central Banks and Ministries of Economy and Finance of different countries.

The dynamism and depth of the foreign exchange market depend on the behaviour pattern of the players. Modern technology has provided full information to players and how they decipher the market information and use; their response determines the course of market movement.

Financial market does not refer to a physical location. Market participants are linked by formal trading rules and communication networks for originating and trading financial securities link market participants. In an environment of deregulated and liberalized financial markets, international capital movements have been driven mainly by economic fundamentals. The macroeconomic conditions of various countries and their trade, monetary and fiscal policies are important variants. Interactions among markets have been facilitated by technological innovations, which provided market participants with opportunities to diversify, hedge, and increase profits on their investments, thereby promoting the use of new financial products and instruments.

Features of International Financial Markets: Let us consider some of the important foreign exchange market features. They are as under:

- i) In a business setting, there is a fundamental difference between making a payment in the domestic market and making a payment abroad. In a domestic transaction, companies use only one currency. In a foreign transaction, companies can use two or more currencies.
- ii) This market is the largest asset class in the world because of its enormous volume of trading, representing high liquidity.
- iii) It spans over a large geographical area.
- iv) It is around the clock markets as it has continuous operation: 24 hours a day throughout the week (Monday to Friday at each centre). i.e., trading from 22:00 GMT on Sunday (Sydney) until 22:00 GMT Friday (New York).
- v) The variety of factors is responsible for fluctuations in exchange rates.
- vi) As compared to other financial markets of fixed income instruments, this market operates on the low margins of relative profit.
- vii) Important foreign exchange trading centers are located in Hong Kong, Singapore, Paris and Frankfurt, amongst others, while the biggest three are New York, Tokyo and London, of which London is the largest.

Cross-border transactions occur through an exchange of currencies between buyers and sellers. The tendency of each country to prefer to use its own unique currency complicates international business transactions. There are around 175 currencies in use around the world. International business would be impossible without foreign exchange and the foreign exchange market. The foreign exchange market refers to the organizational setting within which businesses, governments, and banks buy and sell foreign currencies and other debt instruments. Only a small fraction of daily transactions in foreign exchange actually involve the trading of currency. Most foreign-exchange transactions involve the transfer of electronic balances between commercial banks or foreign exchange dealers. The foreign exchange market has two major segments: the over-the-counter market (OTC) and the exchange-traded market. The OTC market is composed of commercial banks, investment banks and other financial institutions. The exchange-traded markets are composed of securities exchanges, such as London International Financial Futures and Options Exchange (LIFFE), where certain types of foreign-exchange instruments, such as exchange-traded futures and options, are traded.

The foreign exchange market serves two main functions:

- i) To convert the currency of one country into the currency of another.
- ii) To provide some insurance against foreign exchange risk, which means the adverse consequences of unpredictable changes in exchange rates.

A given foreign-exchange market operates at three levels:

- i) in transactions between commercial banks and their commercial customers, who are the ultimate demanders and suppliers of foreign exchange;
- ii) in the domestic interbank market conducted through brokers; and
- iii) in active trading in foreign exchange with banks overseas.

These foreign exchange markets are very important for international businesses. There are four main uses of foreign exchange markets in that sense:

- i) The payments a company receives for its exports, the income it receives from foreign investments or the income it receives from licensing agreements with foreign firms may be in foreign currencies.
- ii) Businesses use foreign exchange markets when they are required to pay a foreign company for its products or services in its country's currency.
- iii) When businesses have spare cash that they wish to invest for short terms in money markets.
- iv) Currency speculation is another use of foreign exchange markets. Currency speculation typically involves the short-term movement of funds from one currency to another in the hopes of profiting from shifts in exchange rates.

In this market currencies of various countries are bought and sold against each other. This is also called foreign currency market where an over-the-counter trade takes place. This market has two wings wholesale market and retail market. In retail market travelers and tourists exchange one currency for another in the form of currency notes or travelers' cheques.

11.3.1 Concept of International Financial Instruments

Foreign exchange instruments are financial instruments that are represented on the foreign market. It is a real or virtual document representing a legal agreement involving some sort of monetary value. Foreign exchange instruments primarily consist of currency agreements and derivatives. Financial instruments can be generally classified as:

- i) Equity based, representing ownership of the asset,
- ii) Debt based, representing a loan made by an investor, to the owner of the asset, and
- iii) Foreign exchange instruments are a unique type of instrument

Different subcategories of each instrument type exist. Financial instruments can be considered as easily tradable packages of capital, each having unique characteristics and structure. The wide array of financial instruments in the contemporary marketplace allows for the efficient flow of capital amongst the world's investors. Any financial instrument has three major functions:

- i) Financial instruments act as a means of payment
- ii) Financial instruments act as stores of value. They generate increases in wealth that are larger than from holding money and can be used to transfer purchasing power into the future.
- iii) Financial instruments allow for the transfer of risk. Futures and insurance contracts enable a person to transfer risk to another. Information about financial instruments is important to users, as it provides insight into the risks related to financial assets and financial liabilities, including the exposure to risks arising from these to the entity and how they are managed.

Check Your Progress 1

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

- 1) Write about globalization of financial markets in around 100 words.

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2) What are international financial markets?

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.....

3) Explain the concept of international financial instruments.

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11.4 TYPES OF INTERNATIONAL FINANCIAL MARKETS

The international financial markets have primary and secondary markets dealing with issue of new instruments and trading in existing instruments and negotiable debt instruments, respectively. International financial markets can develop anywhere, provided the local regulations permit the market and that the potential users are attracted to it. The most important international financial centres are London, Tokyo and New York.

Volume of Foreign Exchange Market: According to the Bank for International Settlements (BIS), trading in foreign exchange markets averaged \$6.6 trillion per day in April 2019. This is composed of the following:

- i) \$2 trillion in spot transactions
- ii) \$1 trillion in outright forwards
- iii) \$3.2 trillion in foreign exchange swaps
- iv) \$108 billion currency swaps
- v) \$294 billion in options and other products

Source: Investor Tonight (2021)

There are different types of financial markets. However, market are classified according to the nature of financial operations:

- i) money markets (or exchange markets): exchange or exchange related transactions,
- ii) credit markets : deposit taking and lending,
- iii) capital markets : issuance of securities, and
- iv) equity markets : issuance of international equities.

A brief description of these markets is as under:

International Currency Market: The main participants in international financial markets are commercial banks, pension funds, hedge funds and private equity funds. An important aspect of growth has been the emergence of international

banking consortia. Since the 1960s various banks started forming international syndicates. Multinational banks are participating in huge transfers of capital not only for investment purposes but also for hedging and speculating against exchange rate changes.

Euro Currency Market: The transactions in Eurodollar, Euroyen, Eurofranc etc. are known as 'Euro Currencies Markets'. This market allows for more convenient borrowing, which improves the international flow of capital for trade and investment between countries and companies. For example, an Indian company borrowing U.S. dollars from a bank in Germany is using the Euro market. The Euro market is a wholesale market (generally restricted to transactions over US \$ 1 million), with participants limited to banks, financial institutions, institutional investors, major corporates and high net worth individuals.

The prefix 'Euro' creates some confusion because it denotes a currency used for financial transactions outside the country of origin of that currency. As for example, if US dollar is called Eurodollar, it means financial assets and liabilities (denominated in dollars) but traded outside the United States. Similarly, Japanese yen traded out of Japan is called Euroyen, and Swiss francs traded out of Switzerland is called Eurofranc. But after launching of the 'Euro' in 2000 as the official currency of European Union, Eurocurrency market is the market in the currency of the Euroland. Further, it is known as Eurodollar, Euroyen, Eurofrancs etc. depending on which particular currency is used for financial transaction outside the country of the origin of that currency.

Euro currency markets consist of banks (Euro banks) that offer wholesale deposits and loans in various currencies. Euro banks are commercial banks dealing in Euro markets. Euro banks are financial intermediaries that simultaneously bid for deposits and make loans in a currency or currencies, other than that of the country in which they are located. Operations of Euro markets have spread over Asia, Europe and the United States.

Euro currency markets facilitate hedging possibilities for corporate borrowers. Euro market is very deep in the short-term market, where 3-6 months deposits are most popular. The basic structure in Euro market operations has been medium to long term lending on variable (floating) interest rates, with an option to re-set (roll over), the interest rate periodically at 3 or 6 months. The banks take into consideration borrowers requirements and have devised various instruments for preferred maturity.

London Inter-bank Bid Rate (LIBID): This is the rate bid by banks on Eurocurrency deposits.

London Inter-bank Offer Rate (LIBOR): This is the rate of interest at which banks borrow funds, in marketable size, from other banks in the London inter-bank market. This is the most widely used benchmark or reference rate for short-term interest rates. A large volume of Eurodollar lending involves the London

Interbank Offered Rate. The LIBOR rate measures the cost to a bank of raising new funds from the market to re-lend. This is the basis of almost all variable (or floating) rate lending in the Euro markets.

When two top-rated banks arrange an interbank transaction, the interest rate agreed for a loan and deposit will often be somewhere between the LIBOR and the LIBID, and possibly the average of these two rates, referred to as LIMEAN as a reference rate for their interbank transactions. The LIBOR convention has spread in various variations. In all cases the concept attempts to measure a bank's cost for a loan.

11.5 IMPORTANCE OF INTERNATIONAL FINANCIAL MARKETS AND INSTRUMENTS

Operations in financial markets have become highly globalized and their importance in contemporary world is evident from the following:

- i) *Maintaining stability in international financial markets is not easy as it has become very intricate.* With easy movement of funds from one country to another, the prices of financial instruments have become highly volatile. Exchange rates, prices of bond and securities are affected by different kind of “news” which make expectations very precarious. Exchange rates and securities prices interact with each other. The degree of interaction differs from one situation to another, depending on a variety of circumstances.
- ii) The information revolution has completely changed economic, financial, and political conditions around the world because of a constant flow of news which result in investors making abrupt changes in their holdings. This has brought larger swings making the market vulnerable both in the short and medium term in a world of integrated financial markets and enormous worldwide liquidity.
- iii) Exchange rates are in a constant flux. Transactions have become increasingly complex.
- iv) International financial markets are very well integrated now as there is no exchange control which makes freer movement of capital across countries.
- v) Cross-listing of securities in several countries.
- vi) Cross-country hedging and portfolio diversification.
- vii) 24-hour trading in financial instruments at exchanges around the world. Automated trading execution systems provide a 24-hour trading market, allowing traders to enter buy and sell orders that are automatically matched according to price and time preferences.
- viii) A shift from banks to nonbank financial intermediaries, such as brokerage houses, securities firms, insurance companies, and pension funds. In such a competitive environment, banking activities have also significantly changed.

- ix) A shift from loans to securities and a rise in the use of foreign financial centers. Surge in the use of new financial instruments such as derivative products: options, futures, and swaps on interest rates, foreign currencies, stocks, bonds, and commodities.

Multinational corporations (MNCs) which operate on a global scale need worldwide sources of finance to meet their investment needs. To serve MNCs, financial institutions have diversified their services such as transactions in foreign exchange, money market instruments, and derivative products worldwide. These sophisticated financial instruments provide investors an array of alternatives for hedging and shifting risks, leading to greater certainty of international receipts and payments. Of course, all these services entail costs. Given the contemporary environment of volatile exchange rates, interest rates, and commodity prices, a large market of different financial instruments exist to help international businesses, speculators, and investors. The growth of new financial instruments has provided flexibility to portfolio management operations. As such, the growth of international financial markets has facilitated cross-country flows which contribute to a more efficient allocation of resources.

Check Your Progress 2

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

- 1) Enlist different types of international financial markets?

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- 2) Who are the players in the international financial market?

.....
.....

- 3) Highlight the importance of international financial markets and instruments.

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11.6 INSTRUMENTS OF INTERNATIONAL FINANCIAL MARKETS

Funds are raised from the international financial market through the sale of securities, such as international equities or Euro equities, Euro bonds, medium-term and short-term Euro notes and Euro commercial papers. In international financial markets all kinds of instruments are denominated in foreign currencies. While the choice of raising finance through various instruments is large

comprising loans, Euro notes and bond, all markets may not be accessible to all borrowers. We will discuss the derivatives at length. First, let us understand a little about Foreign Exchange Markets.

11.6.1 Financial Instruments

Euro is the currency of Euro land and transaction in the Euro currency are known as Euro currency market. These markets thrive as they are outside the control of any central bank. Consequently, Euro banks can offer higher yields to depositors and fixed rates to borrower which makes Euro markets attractive. The important instruments that are used to raise finance are Euro notes, Euro-commercial papers, Medium-term Notes, Floating Rate Notes, Euro bonds and Euro-equities.

Similar to domestic financial markets, international financial markets may be divided into money and capital markets. Money markets deal with assets created or traded with relatively short maturity. The maturity periods of different instruments range from one day to one year. Such instruments are very liquid and relatively risk free. Internationally traded instruments of borrowing and lending of funds of short-term nature are:

- i) treasury bills,
- ii) bank certificates of deposits,
- iii) commercial paper,
- iv) bankers' acceptances,
- v) repurchase agreements, and
- vi) other short-term asset-backed claims.

From the liquidity angle this market is important for MNC's operations as they get fund on day-to-day basis. Treasury bills make up the bulk of the money market instruments. The Capital markets deal with instruments whose maturity exceeds one year.

Derivative Instruments: These financial instruments have values determined from underlying assets, such as resources, currency, bonds, stocks, and stock indexes. These products are very popular and the value of derivative instruments depends on other, more basic, underlying variables. These instruments have been developed to meet the needs and preferences of different customers, including their desire to hedge risks in an environment of fluctuating exchange rates, interest rates, stock prices, and commodity prices. Most of the derivatives except the interest rate swaps are short term in nature.

Options and futures are widespread and traded actively. The five most common examples of derivatives instruments are synthetic agreements, forwards, futures, options, and swaps. Financial institutions, banks and their corporate clients regularly trade in forward contracts, swaps and different types of options outside

exchanges also known as over-the-counter markets. This highlights the fact that there is no single marketplace or an organized exchange for this.

Spot purchase and sale transactions; The spot foreign exchange purchase refers to the business where the designated outlet buys foreign exchange from institutional customers based on the spot exchange rate in the foreign exchange purchase and sale market.

Forward Foreign Exchange Purchase and Sale: The forward foreign exchange purchase and sale refers to the business where the customer signs a forward foreign exchange purchase and sale agreement with the Bank, specifying the foreign exchange currency, amount, terms and exchange rate of a future foreign exchange purchase/sale transaction. This is put in place in order to handle the foreign exchange purchase and sale of the currency. The amount and exchange rate are specified in the agreement at the time provided in the contract. This product enables the customer to preserve the asset value and avoid exchange rate risks by locking up the future exchange rate at present (i.e., the future cost or gains).

The forward rate is based on the prevailing spot rate plus (or minus) a premium (or discount) that is determined by the interest rate differential on comparable securities between the two countries involved. For example, if interest rates in Switzerland are higher than those of the United States, the Swiss franc shows a forward discount, which means the forward rate is less than the spot rate. Conversely, when Switzerland's interest rates are lower than those of the United States, the Swiss franc shows a forward premium, which means the forward rate is higher than the spot rate.

Call Option Vs Put Option: A call option gives its holder the right but not the obligation, to buy the specified amount or quantity of a currency usually against the U.S.A at the strike price. On the other hand, a put option gives its holder the right to U.S.A.

Future: A future is a derivative transaction that exchanges derivatives on a determined future date at a predetermined exchange rate.

Interest Rate Swap: An interest rate swap is a derivative agreement between two parties that involves swapping interest rates where each party agrees to pay other interest rates on their loans in different currencies.

Certificate of Deposit (CD): This is a savings certificate generally issued by commercial banks on which the bearer gets interest. A CD indicates three things (i) a maturity date, (ii) a specified interest rate, and (iii) amount. The amount of CD is not fixed and can be issued in any denomination.

Banker's Acceptance: This is a short-term credit instruments created by a non-financial firm and guaranteed by a bank. The important feature of such acceptance is that it is traded at a discount from face value on the secondary market.

11.6.2 International Debt Instruments

Borrowers are the issuers of debt instruments in the form of Promissory Notes, Bonds and Commercial Papers. There are various classes of borrowers (e.g. companies, governments and other entities) having diversified needs for funds in terms of:

- i) Magnitude of amounts.
- ii) The period for which borrowings are needed. Borrowers may need short term or long term funds for capital expenditure. Governments are known as sovereign borrowers who raise long-term debt in the international market to finance infrastructure and other developmental needs, etc.
- iii) The currency in which borrowings are to be raised.
- iv) The decision for debt finance depends on the costs, borrowing terms and covenants imposed by lenders, project size, etc.

Let us now examine some common debt instruments used for short, medium and long term.

Euro commercial Paper: This is an unsecured, short-term paper issued by a bank or corporation for financing accounts receivables and inventories in the international money market, denominated in a currency other than the corporation's domestic currency. It is usually issued at a discount reflecting prevailing market interest rates. Normally, commercial papers have a maximum maturity of 270 days. The most prominent markets for commercial paper are the United States, Canada, UK, Japan and Australia. Euro market prominence for US \$ and Euro commercial paper are more popular in Singapore and the UK.

Commercial Paper (CP) is marketed either directly by the issuer or through a dealer i.e. investment bank. A large majority of issuers market their CP directly. In the case of dealer-placed CP, the issuers use the services of a securities firm or an investment bank. CPs sold in this way is called dealer paper. Competitive pressures have forced reductions in the fee charged by dealers. CPs are issued at a discount. In other words, if an investor buys a CP with face value of US\$ 1 million, he pays less than the face value, though he receives back on maturity the total face value. The difference between the two represents 'interest'. The difference between the purchase price and the redemption amount makes up the return to an investor.

Bonds: Supranational agencies like the World Bank and Asian Development Bank raise bonds in the international market. National Governments issue government bonds. Municipal or other local authorities issue municipal bonds, while companies issue corporate bonds. Bonds do not have one common interest rate. Rates differ in terms of fixed-rate, floating-rate and convertible bonds. The World Bank issued the global bonds for the first time in 1989. Since 1992, such bonds are being issued also by companies. Presently, there are seven currencies

in which such bonds are denominated: the Australian dollar, Canadian dollar, Japanese yen, DM, Finnish markka, Swedish krona and Euro. The special features of the global bonds are:

- i) They carry high ratings
- ii) They are normally large in size
- iii) They are offered for simultaneous placement in different countries
- iv) They are traded on “home market” basis in different regions.

International bond market is defined as a market for bonds which are sold anywhere in the world but not in the geographical territory of the country of currency in which it is denominated. Mostly the funds raised by the borrowers in such market consist of foreign currencies i.e., other than the borrowers own currency for example, a U.S. company borrowing Euro – Dollars, outside U.S.

The bonds are generally denominated in amounts of US \$ 100,000, \$ 500,000 or more. These are debt securities, normally having a maturity period of one to 10 years. The ‘notes’ mature in less than one year and bonds have maturity up to 10 years. *It is to be noted that often the terms ‘notes’ and ‘bonds’ are used interchangeably.*

Equity financing in the International Markets: Compared to debt financing instruments, issuance of Euro equities stands out as a distinct source of raising finance. Through equity issues, the issuing company offers ownership rights as distinguished from creditorship rights in the form of debt. In recent times there has been increase in cross border equity investment. This global trend is evident from the behaviour of institutional investors of developed countries’ markets, increasing their exposure to equity from emerging markets. Shares of these overseas firms are often traded in stock exchanges like New York and London. The *modus operandi* is that such shares of overseas firms are traded indirectly in the form of ‘depository receipts’. A depository holds the shares issued by the firm in the form of ‘depository receipts’.

Indian companies have been raising finance through Euro-Bonds and Euro-Equities and have been floating ADR and GDR. In the case of US markets, this type of issue is known as ‘American Depository Receipts (ADRs), while Global Depository Receipts (GDRs) is an instrument to raise equity capital in multiple markets outside the issuer’s domestic market through one security - which is traded in a foreign stock market.

11.7 FOREIGN EXCHANGE EXPOSURE/RISK

It is very common to encounter currency exchange rate movements regularly. These unanticipated shifts present risks to the international business because they represent unplanned for changes in the value of asset and liabilities. These risks are usually categorised into three types: transactions exposure, translation

exposure and economic exposure. These exposures result in positions that are either uncovered or covered, that is, hedged. Risk exposures should be determined according to several possible degrees of severity of circumstances. These risks depend on the nature of the item that is exposed, measurability of the exposure and the timing of estimation of exposure. Let us understand various types of risks:

- i) *Transaction risk* arises from foreign currency denominated transactions which an entity is committed to complete. In other words, it arises from contractual foreign currency, future cash flows, when export prices are fixed in foreign currency, or imports invoiced in foreign currencies. The exposure is due to currency exchange rate fluctuations between the time the commitment is made and the time it is payable. There are approaches to dealing with eliminating the risk of a transaction exposure or hedging at the operations level.
- ii) *Translation Risk*: Translation Exposure is concerned with the present measurement of past events. It arises from the need to convert values of assets and liabilities denominated in a foreign currency into the domestic currency. Suppose a company has a different proportion of assets and liabilities denominated in particular currencies. In that case, exchange rate movements are likely to have varying effects on the value of these assets and liabilities. It is also referred to as accounting exposure. This fact makes the measurement of translation exposure dependent on the accounting policies followed for the purpose of converting the foreign currency values of assets and liabilities into the domestic currency.

“For reducing translation and transaction exposure, a number of tactics can help firms minimize their transaction and translation exposure. These tactics primarily protect short-term cash flows from adverse changes in exchange rates. In addition to buying forward and using swaps, firms can minimize their foreign exchange exposure by leading and lagging payables and receivables – paying suppliers and collecting payment from customers early or late depending on expected exchange rate movements. A lead strategy involves attempting to collect foreign currency receivables (payments from customers) early when a foreign currency is expected to depreciate and paying foreign currency payables (to suppliers) before they are due when a currency is expected to appreciate. A lag strategy involves delaying collection of foreign currency receivables if that currency is expected to appreciate and delaying payables if the currency is expected to depreciate. Leading and lagging involves accelerating payments from weak-currency to strong-currency countries and delaying inflows from strong-currency to weak-currency countries”(Hill and Jain, 2009).

- iii) *Economic Risk*: Economic risk is referred to as the risk exposure of an investment made in a foreign country due to changes in the business

conditions or adverse effect of macroeconomic factors like government policies or collapse of the current government and significant swing in the exchange rates (Ramachandran and Vaidya, n d). Many factors can cause economic risk like sovereign default on its payments, political unrest and change in the policies made by the government, and unexpected swings in exchange rate. Change in inflation, interest rates, import-export duties, and taxes also impact the exchange rate. Since this directly impacts trade, exchange rates risk seeming to be a significant economic risk.

Economic exposure is concerned with the long-run effect of changes in exchange rates on future prices, sales and costs. This is distinct from transaction exposure, which is concerned with the effect of exchange rate changes on the individual transactions, cost of which are short-term affairs that will be executed within a few weeks or months.

Although economic risk can be mitigated, it cannot be eliminated entirely. Economic risk is the most intricate risk to foresee, and hence mitigating or formulating plans to control the risk is a challenging task. The management of economic exposure draws on hedging and swap contracts. Hedging activities against exchange rate fluctuation will prove worthwhile to mitigate the risk.

Check Your Progress 3

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) What are the different types of international financial market instruments?

.....
.....

2) What do you understand by the Euro – Currency market?

.....
.....

3) What is Global Depository Receipt (GDR)?

.....
.....

4) Explain the concept of financial risk.

.....
.....

5) What are the types of financial risk?

.....
.....

11.8 LET US SUM UP

In this unit, we discussed about foreign exchange market and international financial market instruments. Foreign exchange market refers to instruments denominated in foreign currencies with a maturity of different periods from one day to more than a year. The foreign exchange market is dominated by many central banks, commercial banks, other financial institutions, and nonfinancial institutions, such as corporations and governments. All these are players in the foreign-exchange market. The foreign-exchange market is divided into the over-the-counter market (OTC) and the exchange-traded market. Foreign exchange is money denominated in another nation's currency or group of nations. The exchange rate is the price of a currency. Companies use foreign exchange to settle transactions involving the imports and exports of goods and services, for foreign investments, and to earn money through arbitrage or speculation.

International financial markets and operations comprise buying and selling currencies, deposit taking and lending and issuance of securities. Market segments are classified according to the nature of financial operations. Money markets deal with exchange or exchange-related transaction, credit markets deals with deposit taking and lending. Capital markets and equity markets are concerned with issuance of securities and issuance of international equities respectively.

An option is the right, but not the obligation, to trade foreign currency in the future. Options can be traded OTC or on an exchange. A foreign currency future is an exchange-traded instrument that guarantees a future price for the trading of foreign exchange, but the contracts are for a specific amount and specific maturity date.

Euro market transactions are financial transactions denominated with currency outside the country of its origin. Euro markets consist of Eurobanks that offer wholesale deposits and loans in favourable jurisdiction and in a variety of currencies other than that of the country where the banks are situated. These Euro banks are entirely free of market regulatory controls. Euro market operations have been medium to long-term lending with floating interest rates. Indian companies have been raising funds from international financial markets by issuing Euro bonds, Euro-convertible bonds and Euro equities. 'Global Depository Receipts (GDRs) are used to tap multiple markets using a single instrument.

Risk is widespread in financial markets. The risk in international financial markets is greater than domestic markets. Various kinds of risk are involved in international financial transactions. Such risks are generally categorised into three: transaction risk, translation risk, and economic risk.

11.9 KEY WORDS

Euro banks	The banks which accept deposits and make loan in foreign currencies.
Euro bonds	A bond placed in the countries other than those in whose currency it is denominated.
Euro-Commercial Papers	These are short-term promissory notes issued on an unsecured basis by commercial and financial institutes.
Euro dollar	Dollar deposited in a bank outside its country of origin.
Euro market	Transactions in Eurodollar, Euroyen etc.
Foreign Bond	The bond sold outside the borrower's country and is denominated in the currency of the country of issue.
LIMEAN	The average of LIBID and LIBOR.
London-Interbank Offered rate (LIBOR)	The rate at which funds are offered to a London or by a London bank to another London Bank.

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11.11 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) See section 11.2
- 2) See section 11.3
- 3) See section 11.3.1

Check Your Progress 2

- 1) See section 11.4
- 2) See section 11.4
- 3) See section 11.5

Check Your Progress 3

- 1) See section 11.6
- 2) See section 11.6
- 3) See section 11.6
- 4) See section 11.7
- 5) See section 11.7

UNIT 12 FINANCIAL AND CURRENCY CRISES *

Structure

- 12.0 Objectives
- 12.1 Introduction
- 12.2 Explaining Financial Crisis
- 12.3 Global Financial Crisis 2007
- 12.4 Unfolding of Global Financial Crisis
 - 12.4.1 Structure and the Role of Credit Derivatives
 - 12.4.2 Residential Mortgage-Backed Securities and the bubble burst
 - 12.4.3 How the crisis of 2007 began?
- 12.5 World's most Devastating Financial Crises in History
- 12.6 The Currency Crisis and Its Effects on Financial Markets
- 12.7 Causes of the Financial Crisis of 2008
 - 12.7.1 US Banking system problem
 - 12.7.2 Global Imbalances
 - 12.7.3 Growth of Securitization and the OTD model
 - 12.7.4 Financial Innovation
 - 12.7.5 US Monetary Policy
 - 12.7.6 Political Factors
- 12.8 The Effects of the Crisis on the Macroeconomy
 - 12.8.1 Credit Demand Effects
 - 12.8.2 Credit Supply Effects
 - 12.8.3 Reduction in Corporate Investment and Increase in Unemployment
- 12.9 Initial Policy Response
 - 12.9.1 Monetary policy
 - 12.9.2 Fiscal policy
- 12.10 Let Us Sum Up
- 12.11 Key Words
- 12.12 Some Useful Readings
- 12.13 Answer/Hints to Check Your Progress Exercises

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12.0 OBJECTIVES

After studying this Unit, you should be able to:

- explain the reason behind Financial and currency crisis;
- describe the causes, features, consequences, types, and implications of the Global Financial Crisis;
- analyse the structure and organisation of the Credit Derivative products and their role in the global financial crisis; and
- explain the regulatory responses to the Global Financial Crisis.

12.1 INTRODUCTION

In this unit, we will understand the phenomenon of financial crisis and its origin. You will also get to learn about the way through which the financial crisis affects any economy. Afterwards we will discuss the Global financial crisis of the year 2007 at length by describing its causes, features, consequences, types and implications. We also discuss the structure and organisation of Credit Derivative products and their role in the global financial crisis. Any financial crisis needs to be managed or combatted. To do so there are regulatory responses, which we will discuss in detail.

12.2 EXPLAINING FINANCIAL CRISIS

Financial System consists of financial markets, financial instruments, and financial services. The financial system plays a pivotal role in channelising savings from savers to dissavers. The Merriam-Webster dictionary defines a crisis as an unstable/crucial time or situation in which a decisive change is impending. A financial crisis is when financial asset prices are nosedived in value, firms, businesses, multilateral organisations and consumers cannot pay their debts, and financial institutions experience a shortage in liquidity. A financial crisis is often associated with a bank run where savers lose faith in the banks because of inadequate deposit insurance. A financial crisis is often associated with one or more phenomena: the bursting of a financial bubble, the mortgage crisis, sovereign default, currency crisis, stock market crashes, credit crunches, severe disruptions in financial intermediation and the supply of external financing to various sectors in the economy, large scale government support (in the form of liquidity support and recapitalisation. A financial crisis is sometimes limited to a single country or one segment of the financial services but is more like to spread regionally or globally. Thus, financial crises are typically multidimensional and can be hard to characterise by employing a single indicator.

There are two types of financial crises such as systemic financial crises and non-systematic financial crises. A systemic crisis refers to the risk of the breakdown

of an entire financial system rather than the failure of one isolated part. It denotes the risk of cascading failure in the financial sector caused by the vertical and horizontal integration within the financial system, as during the Great Depression in the 1930s and the global financial crisis in 2008. As a result, it leads to a severe economic downturn. On the contrary, a non-systemic crisis involves only one or a few markets or sectors, like the Savings and Loan Crisis. Apart from these, the other forms of financial crises are currency crises, capital account or balance of payment crises, debt crises and bank crises. A financial crisis occurs when policymakers and finance ministers ignore indirect warnings given by the country's financial market and cannot understand, estimate and forecast the associated risk. Deficiency in financial regulations and under supervision brings instability to the nation's financial markets.

Financial crises are age-old problems. The series starts with debt and inflation crises in ancient Greece and Rome to banking, debt, and currency crises affecting many countries in modern times. Each past crisis is an essential subject for policymakers, social scientists and financial practitioners as it teaches a lesson about how to handle and prevent future crises. Learning from past mistakes and corrective measures not only reduces the incidence and impact of the financial crisis but also enhances understanding of monetary circulations, the impact of internal and external shocks, exchange rate regimes, interest rate and macro linkages etc. – Irving Fisher, Milton Friedman, John Maynard Keynes, Paul Krugman, and John B. Taylor are famous examples.

The literature associated with the financial crisis opines that fundamental factors, such as macroeconomic imbalances in terms of shocks in money supply, output, prices, wages, interest rates, exchange rates, current account deficits, rising fiscal deficits, the balance of payments deficits in terms of volume and receipts of exports and imports, financial openness measures, trade openness indicators, foreign investment openness indicators are the potential factors behind the financial crisis. Financial crises sometimes arrive by irrational factors such as bank runs, contagion and spillovers among financial markets, limits to arbitrage during times of stress, the emergence of asset busts, credit crunches etc.

Even after the structural economic reforms in many countries, such as the advent of globalisation, privatisation, and liberalisation, now the question arises what effects do these structural reforms in the financial sector have on economic growth and vice versa? If the financial and real sectors are interlinked, why could macroeconomists and statisticians not forecast the financial crisis well in advance? Joseph Schumpeter (1911) was perhaps the first economist to examine this nexus. Schumpeter distinguishes between producers and entrepreneurs, where entrepreneurs take risks and go for new combinations in the form of innovation. In Schumpeter's theory of economic development, this unique combination or innovation is the main driver to take the economy from a lower level to a higher equilibrium through bank credit. However, in the modern

banking system, bank credit severely suffers because of a lack of regulatory control and a deposit insurance scheme to protect the banking system. The gross mismatch between short-term saving liabilities of banks for financing long-term investments and uncertainties in the financial markets leads to panic in the bank customers and bank runs.

Different schools of thought have multiple perspectives on the role of finance in economic growth. But it is widely accepted in the literature that the financial system plays a pivotal role in facilitating savings in terms of investment and contributing towards growth. However, this is subject to regulatory control and market discipline.

12.3 GLOBAL FINANCIAL CRISIS 2007

The global financial crisis, which began to manifest itself in July of 2007, emerged as a liquidity crisis, which can be traced back to the US subprime housing market. When interest rates for loans rose between 2004 and 2006, and the prices of houses began to decline in 2007, failures in the repayment of subprime mortgage loans, as well as failures in the repayment of subprime mortgage-backed debts and other securities, heavily increased. Major financial institutions that had borrowed or invested in mortgage-backed securities of non-depository finance companies were forced to declare bankruptcy or were taken over by other banks. The emergence of the global financial crisis was also the result of the doubtful use of high-risk off-balance sheet instruments with zero liquidity and uncertain value on the side of non-depository financial institutions. Mortgage-backed securities and collateralised debt obligations (CDO), traded by commercial banks to distribute credit risk, which turned out to be illiquid for most of the part, heavily accelerated in the year before the financial crisis surfaced.

By contagion, the housing market downturn and subsequent financial market crisis have also affected the broader US real economy. Due to the growing integration of financial markets and industries during the past 20 years, the crisis spread to Europe with dramatic slowdowns occurring across Europe and the recession widening into a global problem. Some of the factors that led to the financial crisis in the US – high-risk, non-banking mortgage loans, low-interest rates, aggressive (subprime) credit expansion, and ignoring of the credit risks of borrowers – created similar problems in Europe.

The US crisis affected almost every part of the world in the new, globalised world of closely interdependent economies. The global financial crisis highlighted the complex international linkages between financial markets and between the US and European economies.

12.4 UNFOLDING OF THE GLOBAL FINANCIAL CRISIS

The financial crisis in 2007 was a non-systemic crisis linked to subprime mortgages or risky loans to homeowners. The US subprime mortgage market delivered a significant shock to the existing architecture for financial services regulation and supervision worldwide. The foundation of the global financial crisis was the housing market bubble which was the result of a political mandate in the US wherein each homeless US citizen was to be provided with a house. The US Federal Reserve bank of St. Louis instructed all the commercial banks in the US territory to provide an easy loan to all US borrowers interested in financing their housing needs. However, commercial banks found difficulty due to the stringent restrictions like asset and income verification of the borrowers before sanctioning home loans to them. Historically, banks approve a loan if the value of the asset possessions is sufficiently higher than the value of the liabilities the borrowers asked for. Hence, potential homebuyers found it difficult to obtain mortgages if they had below-average credit histories, provided small down payments or sought high-payment loans. Lenders denied such mortgage requests unless protected by government insurance or guarantee schemes.

Nevertheless, with the instructions of the government mandate and the circular from the US Federal Reserve Bank, the commercial banks were asked to revise their lending rate at a lower rate to make the loan easier for home loan borrowers despite their low credit history. This is popularly known as 'Sub Prime Lending Based' home loans. Thus, high-risk mortgages became available from lenders who funded mortgages by repackaging them into pools sold to investors in mid 2003s.

A set of new financial products was used to appropriate these risks, with private-level mortgage-backed securities (PMBS) providing most of the funding for subprime mortgages. These instruments are called credit derivative instruments, financial reengineering products, securitisations, and asset-backed securities. Credit derivatives may also be defined as a specific class of financial instruments, the value derived from an underlying market value of the reference asset bank loan or a portfolio of loans, corporate bonds or other debt instruments, trade receivables etc. These instruments are Collateral Bond Obligations, Collateral Debt Obligations, Collateral Mortgage Obligations, Credit Default Swaps, Total Return Swaps, Credit Linked Notes, Residential Mortgage Based Securities, and Commercial Mortgage Based Securities. This enabled more first-time homebuyers to obtain mortgages and homeownership rose. In the next section, we have described a particular structure of Residential Mortgage Based Securities to understand the global financial crisis.

12.4.1 Structure and the Role of Credit Derivatives

To understand the structure and the role of credit derivatives in the subprime mortgage crisis, let's divide the banks into commercial or conventional and modern or investment banks. Commercial banks accept deposits from customers with non-maturing liabilities to finance the borrowers and investments with long-term maturities. However, investment banks manage the depositors' wealth on their behalf. Credit Derivatives (CDs) are off-balance sheet financial contracts that allow the lender to transfer the credit risk of a financial instrument to a third party against the payment of a fee. They are voluminously used in the US commercial banking sector. The banks use them to mitigate credit risk exposure and expand their credit portfolio. A credit derivative involves mainly three participants. First, the borrower needs credit (reference entity). Second, the lender needs insurance for the loan (protection buyer). Third, the third party or the guarantor who wants to profit by covering the risk of a loan default (protection seller / Counterparty). There are two types of credit derivatives. They are Funded and Unfunded credit derivatives. Unfunded CDs are contingent instruments where the protection buyer does not receive any initial payment from the protection seller (counterparty) for covering the credit risk. Under such an agreement, the counterparty pays only when the borrower defaults. The different types of unfunded CDs are Credit Default Swaps (CDS), Credit Default Swap Options (CDSO), Credit Spread Options (CSO), and Total Rate of return Swaps (TRRS).

On the other hand, funded CDs are instruments where the lender is not exposed to credit risk from this counterparty. This is because the counterparty pays the lender an appropriate sum to cover any future loan default. The different types of funded CDs are Credit Linked Notes (CLNs), Collateral Debt Obligations (CDOs), Collateral Mortgage Obligations (CMOS), Collateral Bond Obligations (CBOs), Commercial Mortgage-Backed Securities (CMBS), Residential Mortgage-Backed Securities (RMBS) and Constant Proportion Debt Obligation (CPDO).

12.4.2 Residential Mortgage-Backed Securities and the bubble burst

Providing easy loans to the homeless forced all the commercial banks in the US territory to disburse the home loan to customers despite their weak creditworthiness. The primary incentive before these commercial banks was that they generate these home loan portfolios for the next 15 years to 20 years of maturity so that they can sell the home loan portfolio to investment banks to enjoy the time value of the home loan portfolio without exposing to the credit risk even though home loan portfolio is risky. This is because any of the borrowers can default at any point in time for the next 15 to 20 years on their commitment while paying towards the principal and interest rate of their total loan value. The investment bank purchases this home loan portfolio and creates a

trustee to manage this portfolio. On the full value of this credit portfolio, the trustee also buys the credit protection in the form of insurance with the condition that the protection seller will compensate the amount if any borrower defaults in the credit home loan portfolio during the maturity of the loan. In anticipation, the trustee pays the regular premium to the credit protection seller during the contract's life. These protection sellers are the same investment banks that have created the trusts to manage the credit home loan portfolio. The trustee, also known as a special purpose entity, has taken some amount as reserve and cash account to utilise for the contingency purpose. Based on the remaining amount, otherwise known as asset-backed securities, the trustee has issued different financial instruments such as senior notes, mezzanine notes, junior notes and equity tranches to issue to the investors. There is a hierarchy among these notes, so senior notes provide less return with low risk compared to the mezzanine, junior and equity tranches. The equity tranche is highly risky and provides a higher return. However, it is difficult to find investors for equity tranches in the market; hence, the same investor bank is the investor for equity tranches. These notes are high investment graded by rating agencies such as Standard and Poor, Moodys, and Fitch. A typical structure of RMBS is given as follows.

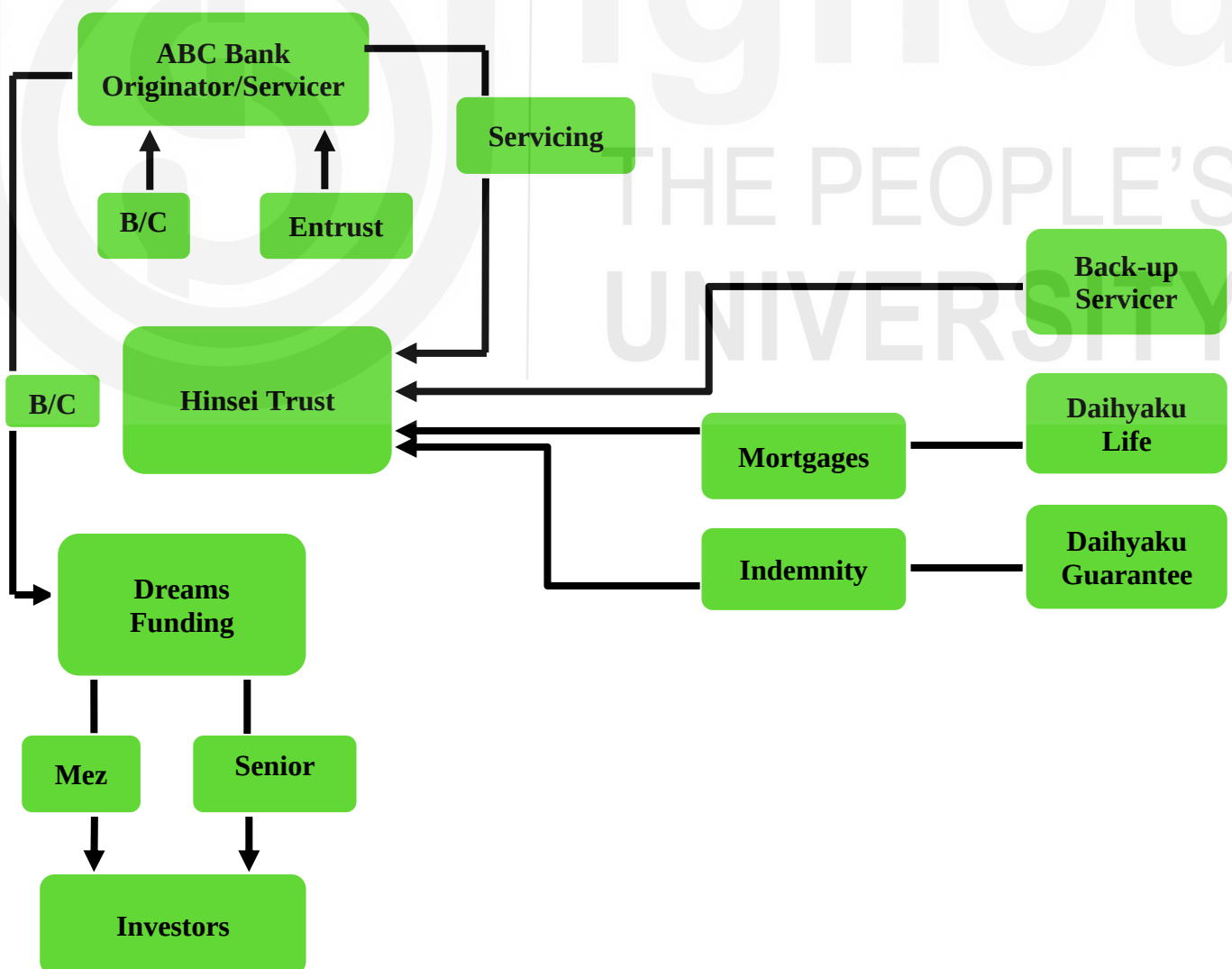


Fig. 12.1: Schematic Diagram of Residential Mortgage-Backed Securities

This structure was followed to finance each US citizen's residential house during 2005-07. This boosted the real estate sector in the US because of the demand for input cost housing sector, employment opportunities in the construction sector and contributed to the US GDP. However, because of the economic cycle and massive unemployment in 2007 in the US economy, the real estate bubble burst in the US economy.. This default among the borrowers and credit protection sellers was so high that Lehman Brothers, one of the pioneer companies in the US economy, declared bankruptcy on 15th September 2008. This is because none of the stakeholders has ever anticipated that each agency will default at the same time. This crisis travelled phenomenally and was declared a global financial crisis. However, because of the conservative banking structure and with a developing financial market structure, the emerging economies were less affected by the global financial crisis..

12.4.3 How the crisis of 2007 began?

The first trouble signals were observed in early 2007 when Freddie Home Loan Mortgage Corporation FHLMC stated that it would no longer acquire high-risk mortgages and that New Century Financial Corporation, a key mortgage lender to risky customers, declared bankruptcy. Another indicator was that the ABX indexes, which track the cost of credit default insurance on instruments backed by residential mortgages, began to reflect increasing default risk assumptions about this time. Most experts agree that the crisis started in August 2007 when short-term funds were significantly removed from previously regarded securities markets. This was evidenced by sharp increases in the "haircuts" on repos and problems faced by asset-backed commercial paper (ABCP) issuers who had trouble rolling over their outstanding paper. It caused stress in the short-term funding markets in the shadow banking system during 2007 was a pervasive decline in US house prices, leading to concerns about subprime mortgages. As previously mentioned, at the start of 2007, these worries were reflected in the ABX index. In the middle of 2007, the credit rating agencies (CRAs) downgraded financial instruments backed by assets. Investors were surprised by the size of the rating actions in mid-2007 regarding the quantity of securities impacted and the average downgrading.

Most economists expected falling housing prices would devastate housing investment and consumption. Very few had forecasted that a crisis in real estate would eventually lead to a mild recession. Even fewer economists had said that this crisis might be more extensive than we think. They had anticipated that this crisis might be the biggest after the Great Depression.

When the financial crisis hit, we realised that the financial system was much more volatile than we had perceived. Within a few months, many banks and financial institutions went bankrupt or were on the verge of bankruptcy. Consequently, banks were unable or reluctant to lend because they feared they might not get their money back, given the economic situation. Investors were unwilling to invest because a lack of confidence in banking led to the failure of

financial intermediaries, and credit availability reached an all-time low. As a result, the interest rate increased dramatically due to the liquidity crunch. The crisis rapidly spread to the world economy, and many banks went bankrupt. Since consumers and firms could not borrow at a very high-interest rate, consumer spending and the output level also fell.

Once the depth of the crisis was analysed, policymakers rushed to respond with financial, monetary, and fiscal policy measures. The central bank adopted a monetary policy by reducing the interest rate. The government responded by adopting expansionary fiscal policies to revive consumer demand to lift the economy out of recession. These policies certainly helped to reduce the severity of the crisis.

Over time, the central bank and the government touched on the limitation of the fiscal and monetary policies. The interest rates touched zero, and we were stuck in a "liquidity trap." Now it was the fiscal policy that was left to lift the economy out of recession. Still, with excessive expansionary fiscal policy and a significant drop in revenues due to lower output, the government had already borrowed a massive amount from the public. A considerable increase in public debt was concerning and worrisome. The government could not do enough because the public debt was too high.

It was 2010 when economic growth turned positive, but the transition to recovery was slow. The unemployment rate was high. Although emerging economies have also suffered from this crisis, large countries like India and China faced a slowdown but not a recession. Massive outflows of funds were seen from these countries because investors did not want to take any risks and were pulling the resources as much as possible to minimise the loss.

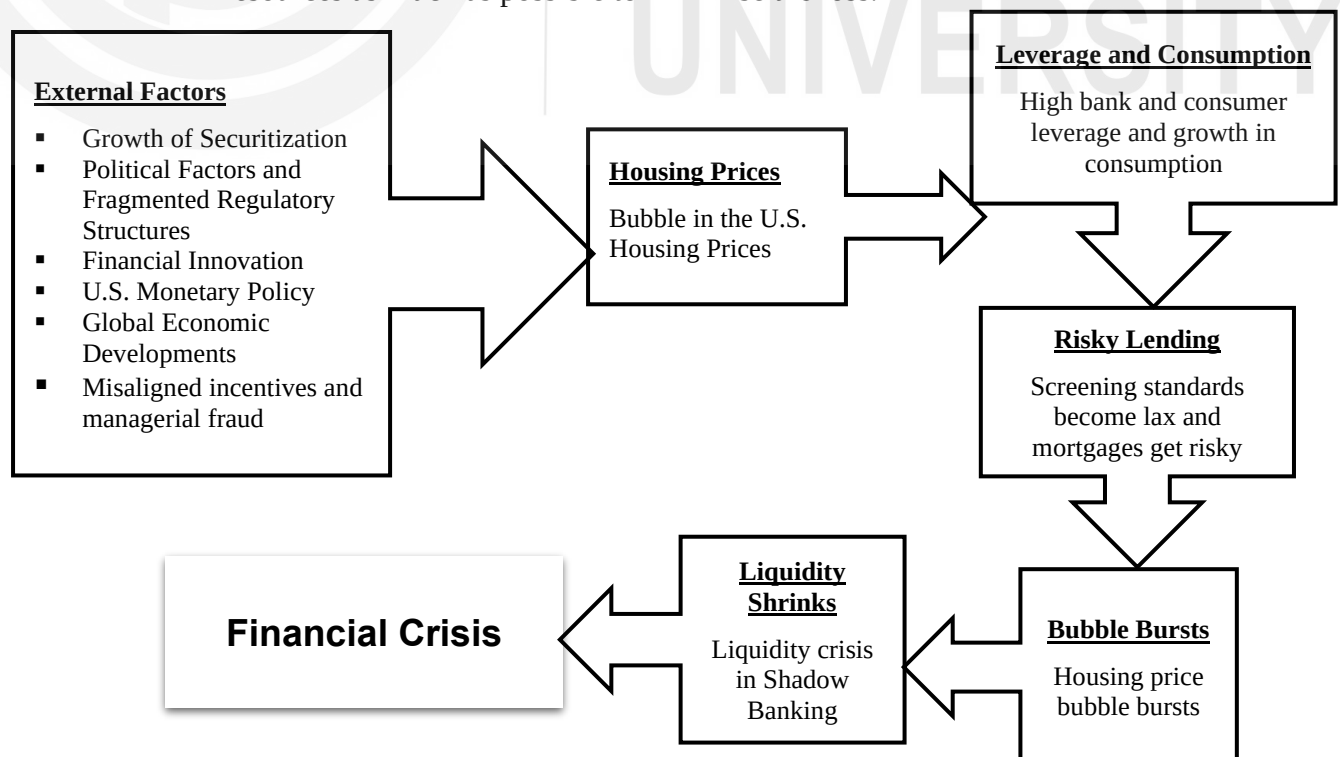


Figure 12.2 Chain of the events leading up to the crisis

12.5 WORLD'S MOST DEVASTATING FINANCIAL CRISES IN HISTORY

Financial crises have occurred throughout history and often lead to the economic tsunamis in the affected economies. Let us have a look at some major crises and their spread.

Table 12.1: Effect of various crisis all over the world

NAME OF THE CRISIS	CAUSE	ORIGIN OF THE CRISIS	AFFECTED COUNTRIES
THE CREDIT CRISIS OF 1772	This crisis was caused by a climate of overconfidence and a time of rapid loan expansion by many British banks, fueled by the vast wealth generated by their colonial holdings and trade.	London	Europe
THE GREAT DEPRESSION OF 1929-39	It was triggered by the Wall Street crash of 1929 and the poor policy decisions of the US government.	United States	Whole World
THE OPEC OIL PRICE SHOCK OF 1973	Retaliation by OPEC member countries against the US for sending arms to Israel in Arab-Israeli War abruptly halted oil exports to the US, and its allies caused significant oil shortages. A severe spike in oil prices caused an economic catastrophe in many wealthy nations, including the United States.	US and developed countries	America, Europe, South Africa, etc.
THE ASIAN CRISIS OF 1997	A period of optimism brought on by speculative capital flows from industrialised nations to the Asian Tigers led to an overextension of credit and an excessive buildup of debt in those economies., and the Thai government abandoning the fixed exchange rate against US dollars, citing a lack of foreign currency created a wave of panic across Asian financial markets.	Thailand	East Asia and Thailand trading partners
THE FINANCIAL CRISIS OF 2007-08	It was triggered by the burst of the housing bubble in the US, which threatened to bankrupt key financial institutions and enterprises.	United States	Whole World

Check Your Progress 1

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) Describe the evolution of financial crises.

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2) How did the Global Financial Crisis of 2007 began?

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3) Briefly discuss the structure and the role of credit derivatives in the subprime mortgage crisis.

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12.6 THE CURRENCY CRISIS AND ITS EFFECTS ON FINANCIAL MARKETS

A currency crisis occurs when the value of a country's currency sharply declines, accompanied by volatile markets and a lack of faith in the country's economy, leading to a negative impact on the economy. As a result, exchange rates become highly unstable, and one unit of one currency may no longer be able to buy as much as it used to in another currency. Currency crises can occasionally be predicted, yet they are often sudden. Many parties can cause a currency crisis, governments, investors, central banks, or any combination of these actors.

Contrary to a currency devaluation as part of a trade war, a currency crisis is an unintended occurrence that should be avoided. Central banks and governments can help stabilise a currency by selling off foreign currency or gold reserves or by intervening in forex markets. Historically, crises have arisen when changes in investor expectations result in a substantial shift in the value of currencies.

Currency crises may be linked to other financial crises, such as abrupt halts in foreign capital inflows, significant increases in capital outflows, and defaults on sovereign debt. A currency crisis is often the symptom, not the cause, of a more severe economic downturn. Central banks are the first line of defense to safeguard a currency's stability. In a fixed exchange rate system, central banks can attempt to maintain the currently fixed exchange rate peg by drawing on the

nation's foreign reserves. They can also intervene in foreign exchange markets when a potential currency crisis is present for a floating-rate system. When confidence in the stability of an economy is generally declining, investors usually try to withdraw money. This is termed as 'capital flight'. Investors convert their domestic currency-denominated investments into foreign currency by selling them. This worsens the exchange rate, resulting in a currency run, making it nearly impossible for the country to finance its capital spending. Currency crisis prediction necessitates examining a diverse and complex set of variables. Few factors that are linked to recent crises are Current account deficit, The value of currencies rose quickly, Investor's uneasiness because of the government's measures.

The Asian Financial Crisis of 1997 is an example of a financial crisis that began as a currency crisis when Bangkok delinked the Thai baht from the US dollar; it sparked a series of currency devaluations and large-scale capital outflows.

12.7 CAUSES OF THE FINANCIAL CRISIS OF 2008

Most economists had forecast the effect of a decline in housing prices would be limited to a decrease in aggregate demand and a slowdown in growth. What they had not predicted was the impact of declining housing prices on the financial system. Sooner falling housing prices crashed the financial system as a whole. Even though it was centered on Wall Street at its peak, the global financial crisis resulted from both the US domestic and international factors.

12.7.1 US Banking system problem

Central banking's inability to force commercial banks to look out for the borrower's creditworthiness before lending and maintain a high capital ratio and low leverage ratio for transparent and robust banking led to the crisis. Commercial banks had overextended credit without accessing the borrowers' creditworthiness. And government's extensive interference and guarantees for the subprime also contributed to this crisis. For efficient and good banking, commercial banks need to maintain a good capital ratio and keep the leverage ratio minimum to show that any sudden volatility can be absorbed. But commercial banks did precisely the opposite; they did not abide by the minimum capital ratio required to maintain as per the financial regulations. If banks had maintained the high capital ratio, they wouldn't have made high returns, so they ignored the capital ratio, and because of that, the leverage ratio increased sharply. Banks probably underestimated the risks they took in good times, just like people.

It all began in the 2000s when interest was meager. Buying a house with a mortgage was easy as the mortgage rates were unusually low. The demand for housing increased significantly, which pushed up housing prices. Lenders saw it as an opportunity to profit and were willing to lend to risky borrowers aggressively. And these risky mortgages are known as subprime mortgages.

By 2006, 20 per cent of all mortgages were subprime in the US. Most economists considered this a positive development. It allowed more people to buy homes by thinking that housing prices would continue to rise relative to the house prices and they would make a profit.

In late 2006, when housing prices started to fall, many borrowers found themselves in a position where the mortgage value exceeded the house's value. It could be easily seen that the mortgages were much riskier than the lenders had assumed, and the borrowers understood. As the housing prices turned around, many borrowers started to default, and banks found themselves in a very tricky position where they were making a considerable loss. It created panic among investors, borrowers, and lenders. Investors started to sell off their financial assets, including securitised mortgages, and even went further by liquidating bank deposits. Due to liquidity shortages, even some banks began to go bankrupt. This spread had infected the other banks too. Banks were reluctant to lend even though banks were not providing liquidity to banks. The whole financial system collapsed.

12.7.2 Global Imbalances

Global trade imbalances were also significant contributors to the financial crisis. United States current account deficit with the oil-exporting countries and the Asia Pacific nations was huge during that period. Especially countries like China, Japan, and other East Asian economies had a huge trade surplus with the United States. It was not that the only United States was going through a trade deficit, but countries like Australia and New Zealand too had a considerable trade deficit.

China was the major manufacturing hub and a favorite destination for investors that helped China accumulate huge dollars reserve arising out of trade surplus and the foreign direct investment inflows. It all was facilitated by China following an undervalued and tightly pegged exchange rate. It assisted the People's Bank of China to accumulate a sizeable foreign currency reserve. Many multinational companies invested there, such as Motorola, LG, Nokia, Toshiba, etc. China has been growing at a miracle rate of 10 percent for the last two decades. By 2008 People's Bank of China had accumulated a foreign exchange reserve of more than \$ US 2 trillion, most of which were government treasury bonds. China and other Asian countries' export-led growth created a huge current account deficit with the developed economies. Asian countries have a huge trade surplus with the United States, and not letting the appreciation of exchange rates lead to substantial capital outflows from the Asian economies to the US economies. And huge capital inflows to the US economy sharply led to a fall in interest rates.

Such low-interest rates made commercial banks consider parking the fund in the US housing sector. Growing China and falling interest rates in the US and global markets can be easily understood by the global loanable fund's framework. A rise

in global saving shifted the saving supply schedule to the right and pushed the United States and World interest rates down.

Extremely low interest rates in the United States incentivised commercial banks to invest in housing. They started to lend without doing a risk assessment, which led to the housing crisis, and the crisis in the housing crisis caused the financial system failure and affected the macroeconomy.

12.7.3 Growth of Securitization and the OTD model

It has been asserted that the US government's intention to broaden ownership was coupled with monetary policy that allowed banks to have softer lending rules. Additionally, the originate-to-distribute (OTD) paradigm of securitisation, a lax monetary policy, and weak supervision over bank capital all contribute to this softening. Due to these circumstances, commercial banks increased their mortgage lending during the time leading up to the crisis, while investment banks engaged in warehouse lending via nonbank mortgage lenders. Additionally, there is empirical proof that the OTD model incentivised banks to make risky loans in ever-increasing numbers.

12.7.4 Financial Innovation

For more than two decades before the financial crisis, we saw an explosion of financial innovation. Information technology played a role, making it more straightforward for banks to create tradable instruments and increasing the interconnectedness of commercial banks with the shadow banking system and financial markets. But apart from information technology, there had to be economic incentives for banks to innovate. Financial markets are incredibly competitive, making it challenging for institutions to have significant profit margins with standard financial products—those whose payout distributions everyone agrees upon. This promotes the hunt for new financial products, particularly ones with contested creditworthiness. The absence of agreement on the investment value of new financial products reduces competition in that market. It enables the institutions supplying those products to make significant initial profits. But since these new items don't have a history, they are also risky. The explanation is that there is a lack of historical data that can be relied upon to resolve the conflict, which means that not only rivals but also financiers of the institutions offering these goods may disagree if they are investments worth making. When this occurs, short-term funding for innovative institutions is not rolled over, resulting in a funding crisis. Before the crisis, the explosion of new asset-backed securities created by securitisation created an ideal environment for this.

12.7.5 US Monetary Policy

Taylor (2009) argues that the price boom and subsequent crash that preceded the crisis greatly influenced the US Federal Reserve's easy-money monetary policy,

particularly in the six or seven years before the crisis. He provides evidence that, between 2007 and 2009, monetary policy was too "loose fitting" in that actual interest rate decisions were significantly lower than what historical experience would indicate should be the basis for policy under the Taylor rule. Taylor (2009) demonstrates how the Federal Reserve's deliberate monetary policy choice, which included these abnormally low-interest rates, fueled the housing boom and ultimately caused the housing crisis. The study uses regression to calculate the empirical relationship between interest rates and housing starts. The results demonstrate a robust positive correlation between the intertemporal decrease in interest rates between 2001 and 2007 and the housing market boom. The impact of low-interest rates on housing prices was amplified by the incentives provided by the low-interest rate environment for lenders to make riskier loans. When the central bank keeps interest rates on loans low for an extended period, it reduces banks' net interest margins, and one way for banks to respond is to increase these margins by taking on more risk. This led banks to expand the borrower pool by lending high-risk borrowers who had previously been excluded, further fueling the housing price boom.

12.7.6 Political Factors

Rajan (2010) argues that structural flaws in the educational system that resulted in unequal access for diverse societal sectors are to blame for the widening economic inequality in the United States. Politicians from both parties saw the expansion of home ownership to deal with this growing wealth inequality—a political proclivity that dates back at least to the 19th century Homestead Act and therefore adopted legislative initiatives and other inducements to force banks to extend mortgage loans to a wider borrower base by relaxing underwriting standards, which resulted in riskier mortgage lending. The elevated demand for houses pushed house prices and led to the housing price bubble.

12.8 THE EFFECTS OF THE CRISIS ON THE MACROECONOMY

The financial crisis immediately affected the macroeconomy two folds. First, the interest rate , at which firms and people could borrow rose a lot. Usually, firms with AAA ratings borrow at a rate close to the government because the AAA rating is considered safest, and bonds with BBB ratings borrow at a higher interest rate. Still, the difference is slight, which is close to 1%. The difference in borrowing rates between the government bonds and the AAA and BBB bonds increased significantly in 2008. Borrowing rates on the government remained very low, whereas the borrowing rate on the AAA and BBB were high. This jump in interest rate made it difficult for the firms to borrow fund. It also became difficult for ordinary people to get a loan because loan default damaged their credit score, and banks were no longer willing to take risks.

Second, the September 2008 financial crisis triggered confusion among consumers and firms. People were confused and anxious about future events and were speculating about another Great Depression. This anxiety among the consumers and the firms led to a drop in confidence. Lower consumer confidence led to lower housing and stock prices, which resulted in a sharp decrease in aggregate demand. The real-world implications of this financial crisis were substantial. These included weaker household loan demand and supply, lower corporate investment, and more unemployment.

12.8.1 Credit Demand Effects

When the housing bubble burst, it jolted household balance sheets, eroding the net worth of households. As a result, the highly leveraged households cut back on their consumption. However, because of various economic frictions related to nominal price rigidities and a lower bound of zero on nominal interest rates, relatively unlevered households did not increase consumption to offset this decline.

The consumption fell sharply in places with higher reliance on home as a source of wealth and in high-leverage counties with more considerable house price reductions. Consequently, consumer spending and credit demand also decreased.

12.8.2 Credit Supply Effects

There is persuasive empirical evidence that the crisis prompted a considerable decrease in the amount of credit banks were willing to extend. One indication is the drop in syndicated loans during the crisis, which is significant because syndicated lending is an essential source of credit for the corporate sector. In addition to banks, institutional investors, hedge funds, mutual funds, insurance companies, and pension funds are also included in the syndicated loan market. The syndicated lending started to decline in the middle of 2007 and accelerated in September 2008. An increase in the cost of credit accompanied the decline in lending volume.

12.8.3 Reduction in Corporate Investment and Increase in Unemployment

It is not unexpected that corporate investment declined and unemployment rose due to a decline in consumer consumption and a rise in the cost and scarcity of credit. In 2008 and 2009, about nine million jobs were lost in the United States, accounting for around 6% of the workforce as the country descended into a deep recession. Additionally, it prevented many from attempting to return to employment after the crisis subsided, which caused the labour participation rate to plummet. This had the effect of making subsequent calculations of the unemployment rate tend to underestimate the actual jobless rate. The recorded unemployment rate increased every month from 6.2% in September 2008 to 7.6% in January 2009. By mid-2009, the average housing price in the United States had

decreased by 30%, and the value of stocks had been reduced by almost 50%. The American auto sector also took significant damage. October 2008 car sales were down 31.9% from September 2008 figures. A causal link between the reduction in credit supply during the crisis and an increase in unemployment. Evidence shows that household access to bank loans tended to matter more than firm access in determining the decline in employment in the manufacturing sector. Still, there was also a significant impact from decreased access to commercial and industrial loans and consumer instalment loans.

12.9 INITIAL POLICY RESPONSE

12.9.1 Monetary policy

The central bank went for overall facilitated liquidity to the financial system to contain the damage. Given the uncertainty, federal deposit insurance was created to prevent confused depositors from rushing to withdraw the deposits. During the crisis, if the investors wanted to take their funds back, the banks had no choice but to sell some of the assets; even sometimes, the bank had to sell at a fire rate, and in the worst scenario, banks went bankrupt. To prevent this, the central bank incorporated several liquidity facilities to meet the bank's short-term fund needs. It allowed the banks to borrow from the major banks. Financial institutions were permitted to use collateral to borrow from the fed.

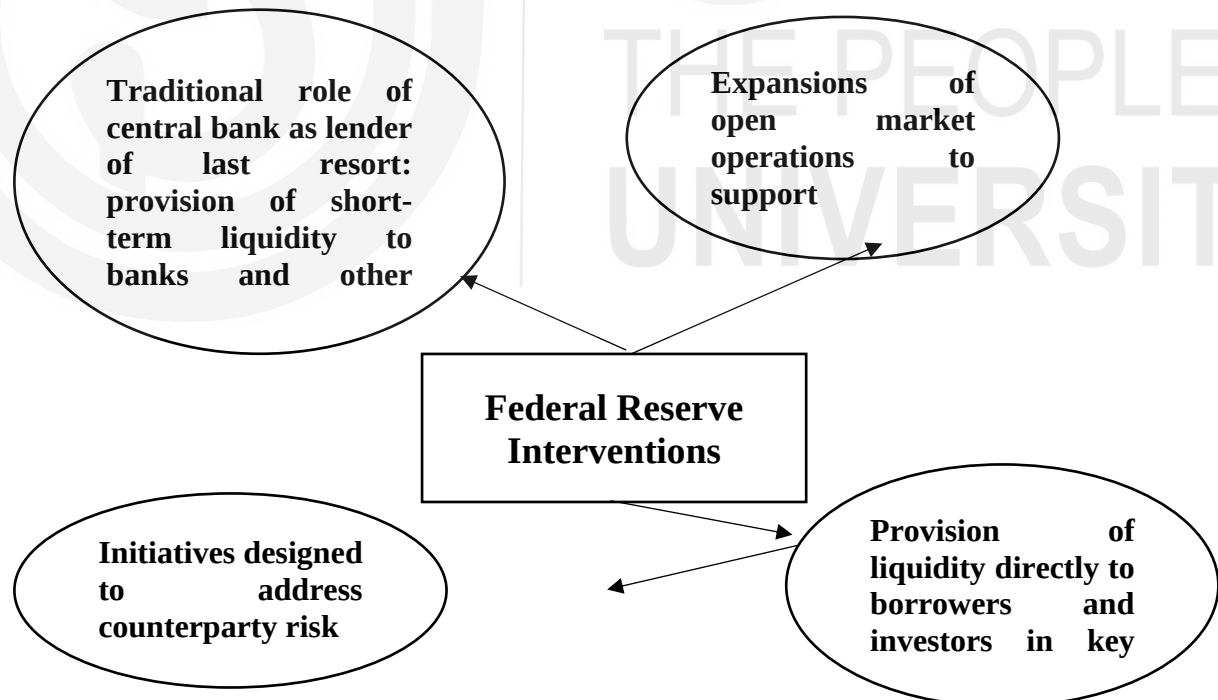


Fig. 12.3: Federal Reserve Interventions

The central banks' facilities provide much-needed liquidity to the financial system. The central banks implemented an expansionary Monetary policy to lower the interest rates so consumers and firms could easily access much-needed liquidity. The monetary authority started to restructure the bank's bad debts, so a

good bank balance sheet brings investors' confidence into the system. During last months of 2009, interest rates reached near zero, and interest rates could not go down anymore. We are stuck in a liquidity trap.

12.9.2 Fiscal policy

Initially, the government introduced the Troubled Asset Relief Program (TARP), which mainly focused on cleaning up the banks. This program's initial target was 4700 billion and was introduced in October 2008. The primary purpose of TARP was to remove the bad assets from the bank's balance sheets, thus reducing uncertainty, reassuring the investors, and making the banks' balance sheets healthier. Assessing the value of impaired assets was extremely tough and time-consuming; they realised it and abandoned the original goal. The new goal was to increase the banks' capital and reduce the leverage ratio. The government started to acquire the bank's shares and provide funds to most of the largest banks to avoid the banks going bankrupt.

Once the adverse shock of the crisis became apparent, the US government adopted an aggressive expansionary fiscal policy. The government reduced taxes and increased spending. After assuming office, the Obama administration prioritised a fiscal program that would increase the demand and reduce the size of the recession. The name of that fiscal program was American Recovery and Reinvestment Act, passed in February 2009. This fiscal program package was \$780 billion in the form of both tax reduction and spending. The government aggressive expansionary fiscal policy caused a large US budget deficit. The deficit increased from 1.7% of GDP in 2007 to 9% in 2010.

Despite the extensive use of a combination of monetary and fiscal policy, output decreased vastly. The economy recovered, but the recovery was slow.

Check Your Progress 2

Note: i) Use the space given below for your answers.

ii) Check your progress with those answers given at the end of the unit.

1) Enlist the causes behind the global financial crisis of 2008.

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2) Explain the macroeconomic effects of the global financial crisis.

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- 3) What are the policy responses initiated by the U S government to combat the global financial crisis of 2008.

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12.10 LET US SUM UP

In this unit we discussed about origin of various financial crises with special focus on Global financial Crisis of 2007. A good institution is a primary requirement for the growth of the economy. It includes well-developed, transparent, and accountable central banking. It appears that the crisis resulted from the interaction of many factors: politics, monetary policy, global economic developments, growth of securitisation, and a fragmented regulatory structure. The central bank must actively regulate commercial banks to follow the guidelines to maintain the minimum capital ratio and limit the leverage ratio to a certain threshold level. It is widely acknowledged that banks will need to operate with more significant capital levels to manage bankruptcy risk and reduce the chance of repeat crises. The shock observation of the economy depends on how well the financial system is developed and how transparent and accountable the country's central banking is. The global financial crisis made policymakers consider the requirement of a tightly monitored financial system and controlled commercial banking to avoid future crises. And the policymakers should adopt a monetary and fiscal policy to reduce the severity of the macro crisis and rescue the economy from recession. During the global financial crisis of 2008, the policymakers did implement a combination of fiscal and monetary policy, but they acted late. Of course, the monetary and fiscal policy had reduced the depth of the recession. Still, if the policymakers had acted upon it early, the recession's severity could have been minimised.

In order to recognise the importance of the central role of banks in the financial crisis and acknowledge the interlinkages of macro and financial sectors of the economy to achieve adequate economic growth with financial stability, the 2022 Sveriges Riksbank Prize in Economic Sciences in Honor of Alfred Nobel awarded to Ben S. Bernanke, Douglas W. Diamond, and Philip H. Dybvig for their research on banks and financial crises. Douglas Diamond and Philip Dybvig developed theoretical models to explore banks' role in the economy and why they are vulnerable to bank runs. Diamond and Dybvig (1983) presented a theory of maturity transformation. They showed that the most efficient arrangement is an institution using demand deposits to finance long-term projects. However, this arrangement has an inherent vulnerability: bank runs may arise. Bernanke (1983) showed that the downturn became so deep and so protracted partly because bank failures destroyed valuable banking relationships. The resulting credit supply contraction left significant scars on the real economy.

12.11 KEY WORDS

Collateralised debt obligations (CDO)

A Collateralized Debt Obligation (CDO) is a type of derivative contract and is a synthetic investment product that groups different types of loans to offer a single investment product to institutional investors. Collateralized debt obligation (CDO) clubs various loans and derives its value from an underlying asset.

Credit derivatives

a type of derivative in which the risk that a loan will not be repaid is sold to a party other than the lender.

Financial Crisis

A financial crisis is when financial asset prices are nosedived in value, firms, businesses, multilateral organisations and consumers cannot pay their debts, and financial institutions experience a shortage in liquidity.

Liquidity trap

A liquidity trap is an adverse economic situation that can occur when consumers and investors hoard cash rather than spending or investing it even when interest rates are low.

Mortgage-backed securities

A mortgage-backed security (MBS) is a type of asset-backed security (an 'instrument') which is secured by a mortgage or collection of mortgages.

OTD model

In the OTD model the originator of a loan sells it to third parties through a securitisation process.

Securitization

Securitization is the process in which certain types of assets are pooled so that they can be repackaged into interest-bearing securities. The interest and principal payments from the assets are passed through to the purchasers of the securities.

Subprime lending

A subprime mortgage is generally a loan that is meant to be offered to prospective borrowers with impaired credit records.

12.12 SOME USEFUL REFERENCES

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12.13 ANSWERS/HINTS TO CHECK YOUR PROGRESS EXERCISES

Check Your Progress 1

- 1) See section 12.2
- 2) See section 12.4.3
- 3) See section 12.4.1

Check Your Progress 2

- 1) See section 12.7
- 2) See section 12.8
- 3) See section 12.9

